Targetboard.ai

Candidate Test — Jira Status “As Of” (Postgres + dbt + SQL)

By: Lior Ash (052-2558662)

Goal

You’ll connect to a remote **Postgres** database containing **raw Jira exports**, build **cleaned dbt models**, and implement a **Postgres SQL function** that returns an issue’s status “as of” a given date.

Details

1. You’ll be using data extracted from Jira using Stitch Data. You can see the documentation of this extractor here
2. The data is saved to a Postgres 17 database. Connection details provided bellow
3. Make sure to review the required deliverables - as we will use them to review your task.
4. You may use any LLM you wish to help out with this task.

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1) Connect to Postgres

Connections credentials

- username = bi
- password = <Will be sent separately>
- host = rb-1-jan-26-backup-do-user-13721082-0.l.db.ondigitalocean.com
- port = 25060
- database = targetboard_production_data_0001
- sslmode = require

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2) Build cleaned Jira models in dbt

You need to create DBT models that **clean** the raw Jira data such that the downstream function uses only cleaned models.

Cleaning requirements

1. Focus on the tables**: issues, changelog changelog__items.** Documentation related to these tables is available on Stitch Data Jira Integration.
2. Keep only issue types**:** `Task`, `Epic`, `Story`
3. **Uppercase issue summaries (**first letter of any word**)**
4. Your cleaned layer must be what your function queries (not raw tables).
5. You **do not need** to clean any jira data - clean only data that is required for step 3

IMPORTANT! Make sure to use dbt_dev_<your first name> when creating the schema with DBT.

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3) Create a Postgres SQL function: status of an item on a specific date

Function requirements

Create a SQL function that:

- Accepts:
  - `p_issue_key text`
  - `p_date date` (interpretation below)
- Returns:
  - `issue_key`
  - `summary`

- `status_as_of` (status value as of that date)

Deliverables

1. **Git repo** including:
   - dbt project with models
   - SQL for the Postgres function (in `sql/` or dbt `macros/`)
2. DBT schema name on the provided database.
3. A short example query demonstrating your function, e.g.:
   - `select * from analytics.issue_status_as_of('ABC-123', '2024-03-01');`
